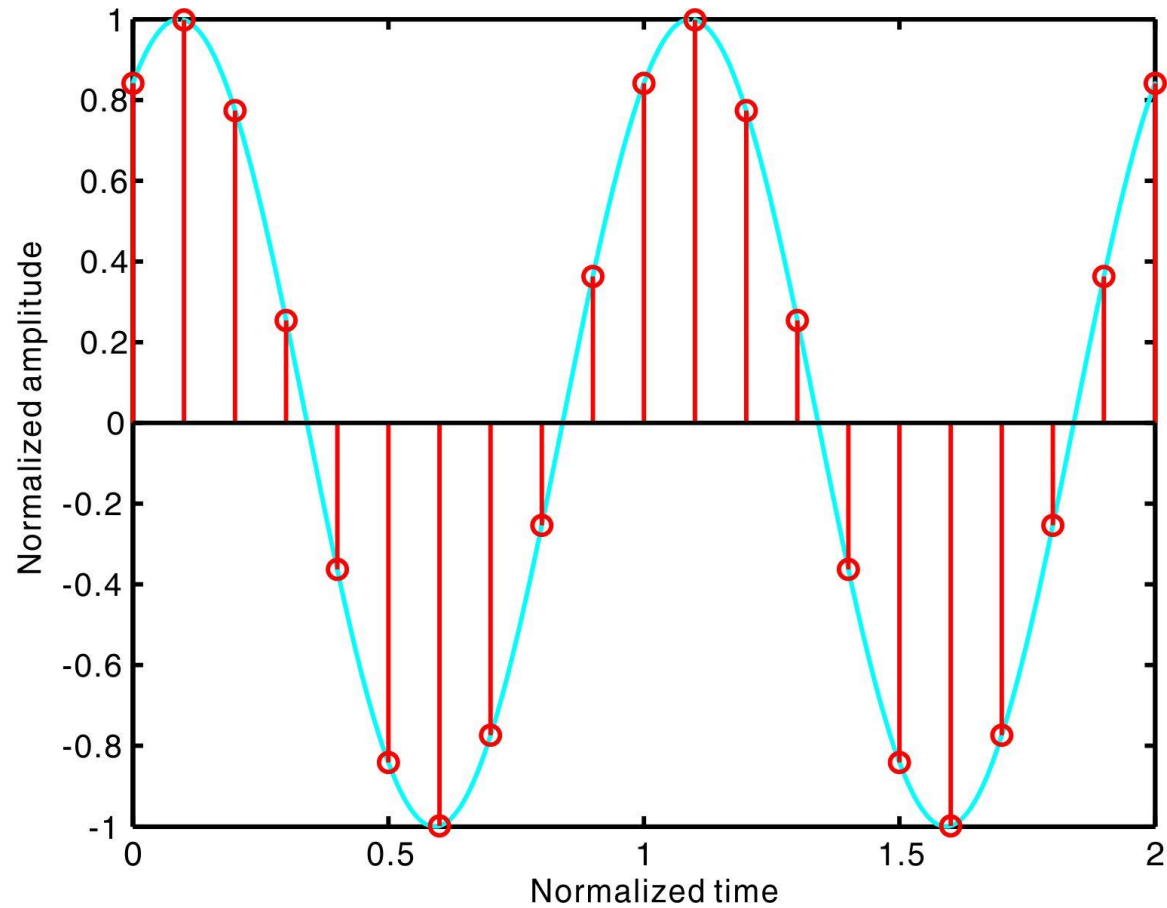


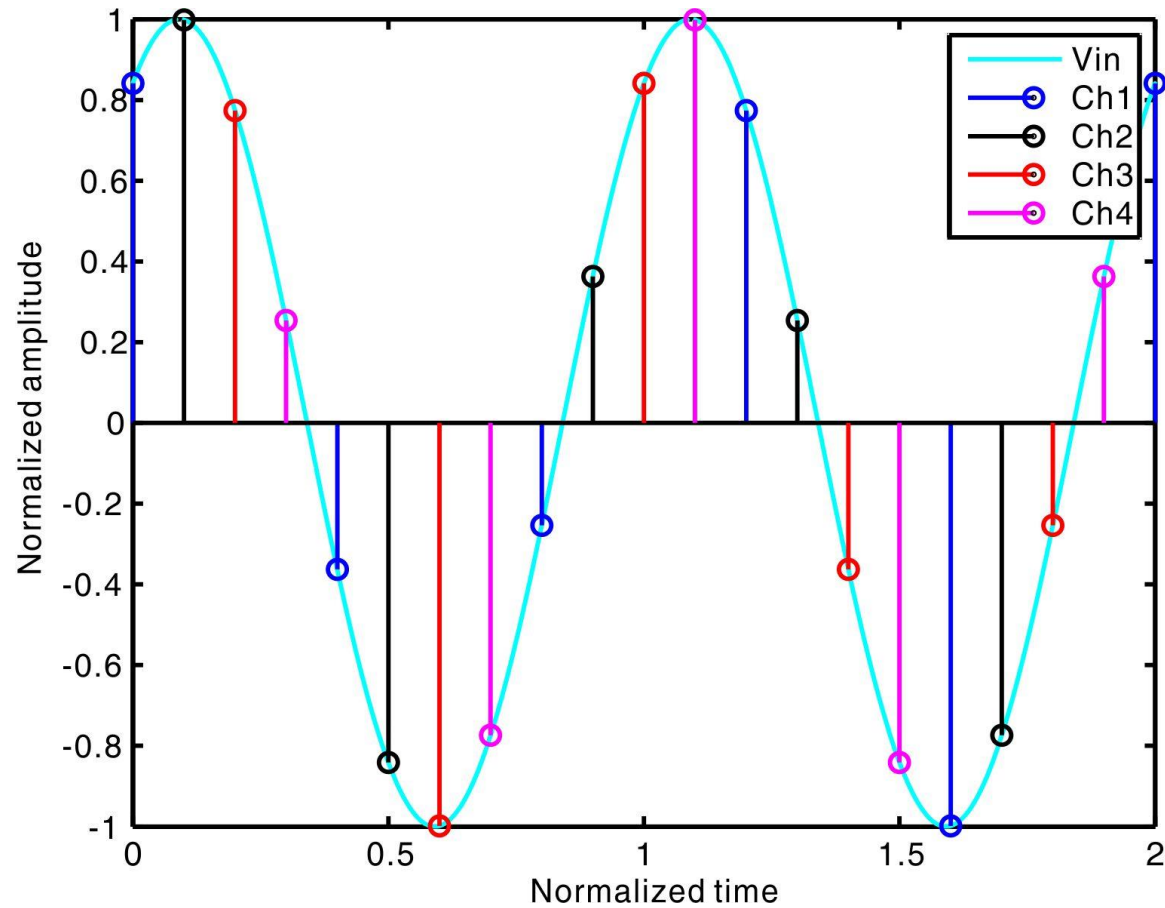
Time Interleaving

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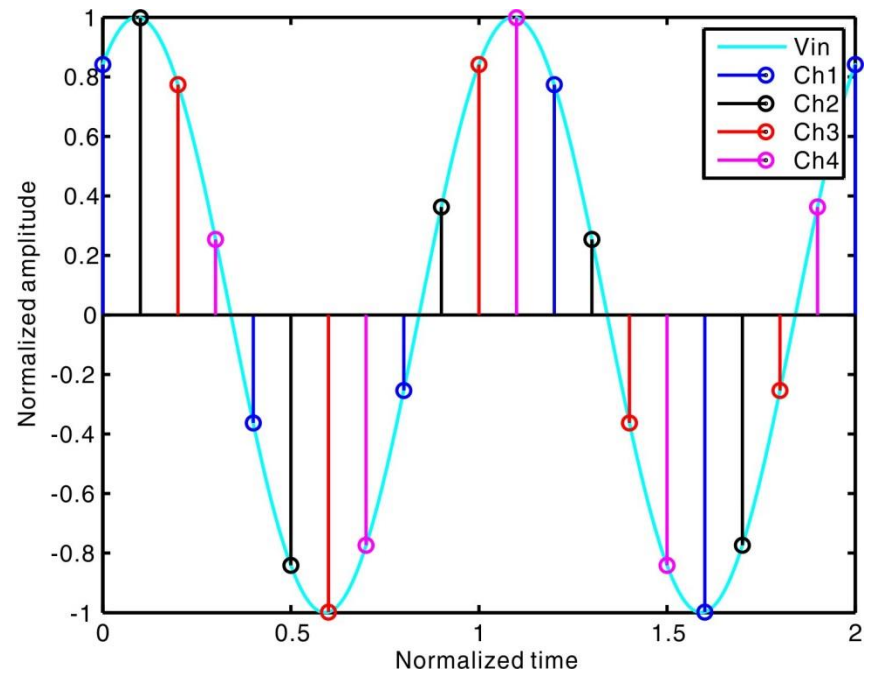
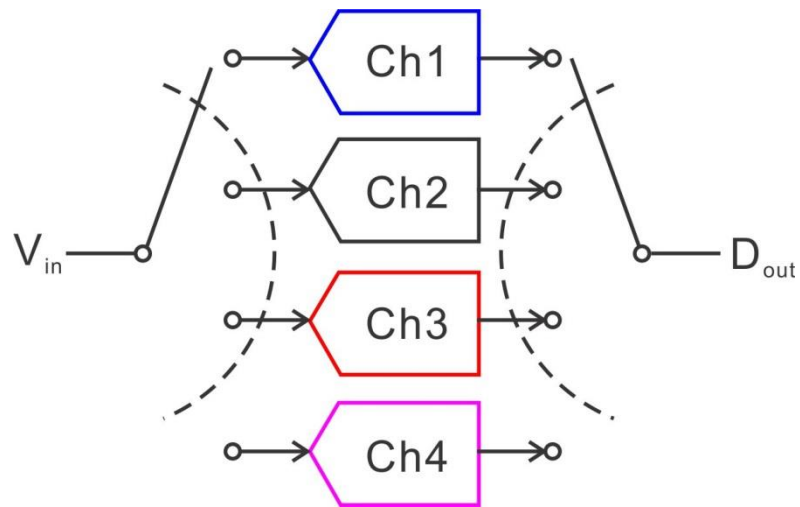
Single Channel ADC



Time Interleaved ADC



Conceptual Implementation



How Fast Can a Pipelined ADC Go? (1)

- Non-dominant pole in two-stage amplifier hard to move past $f_T/5$
- For 73° phase margin (optimum for fast settling), loop crossover frequency is $1/3$ of non-dominant pole frequency
- Settling linearly to 0.1% precision takes 7 loop time constants; typically budget ~ 10 time constants
- Ideally, we'd have $1/2$ clock cycle to settle linearly, but there is some time needed for slewing and non-overlap clock timing
 - Assume 60% of half cycle is available for linear settling
- In summary (here we have assume that the feedback factor is 1)

$$f_{CLK,max} = \frac{f_T}{5} \frac{1}{3} \cdot 2\pi \cdot \frac{1}{10} 0.5 \cdot 0.6 = \frac{f_T}{80}$$

How Fast Can a Pipelined ADC Go? (2)

Technology	NMOS f_T (at moderate $V_{GS}-V_t \sim 150\text{mV}$)	$f_{\text{CLK,max}} = f_T/80$
0.35 μm	10GHz	125 MHz
0.18 μm	30GHz	375 MHz
90nm	90GHz	1.125GHz

- Sampling speeds of 200-300MHz are "easily" achievable in today's technologies
 - f_T is no longer a showstopper
 - Speed ultimately constrained by power and clock jitter

How Fast Can a SAR ADC Go?

- Total conversion time $\approx B(t_{\text{comp}} + t_{\text{Logic}} + t_{\text{DAC}})$
- There are technology limits for all of them
- We cannot trade power linearly with speed

Why Interleaving?

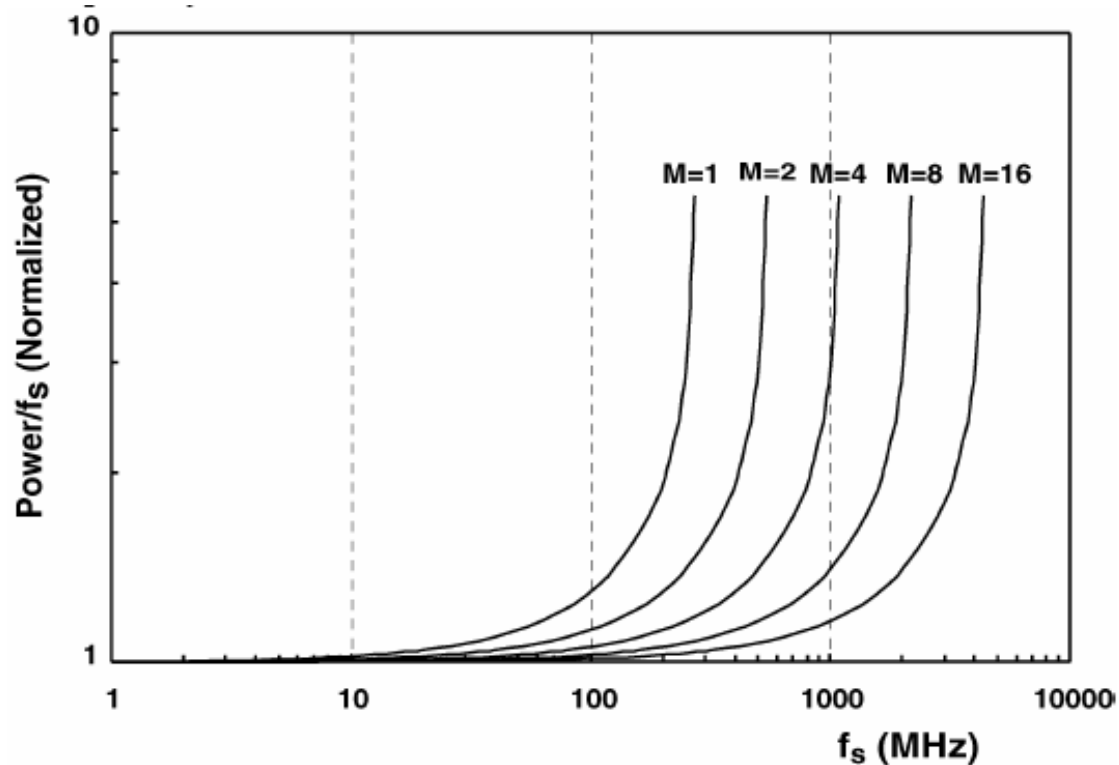
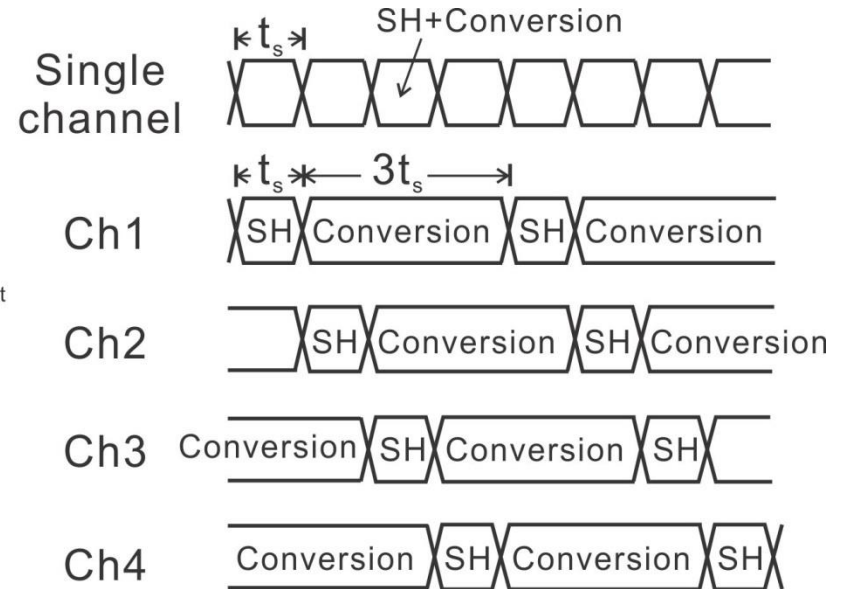
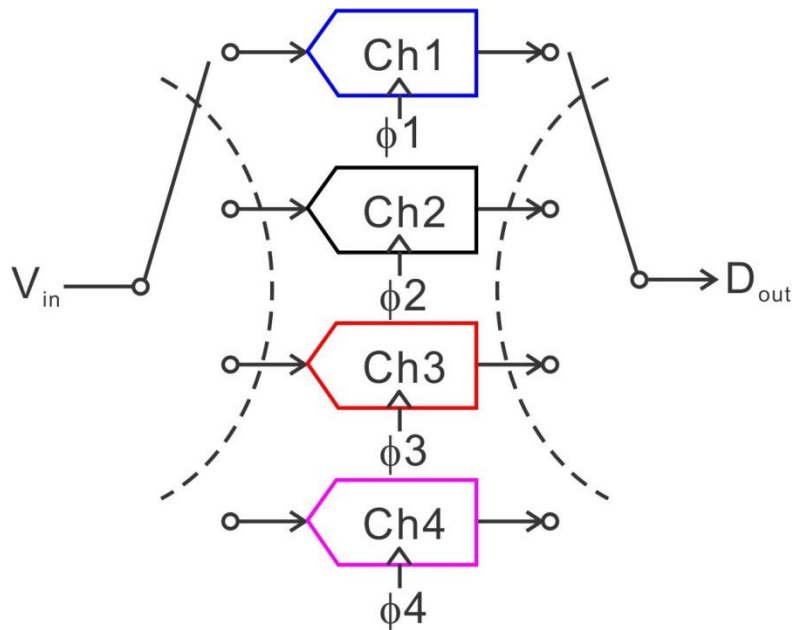


Fig. 13. Power/f_s versus f_s in parallel pipeline ADCs (# of channels (M) is 1, 2, 4, 8 and 16).

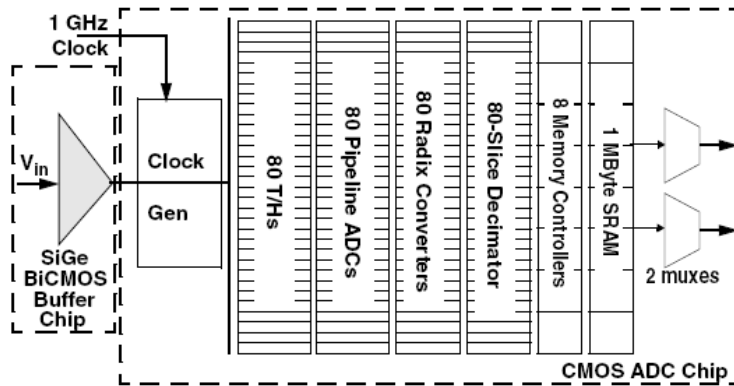
- There are hard frequency boundaries

Time Interleaved ADC

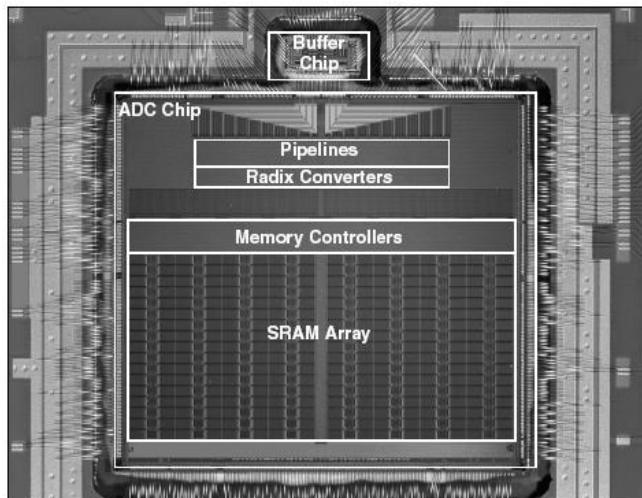


- Idea: interleave 4 ADCs so that each ADC runs as $f_s/4$
- Originally, (SH + Conversion) needs to finish within t_s
- Now, both the SH and conversion speed are relaxed.
 - Note that SH has to finish within t_s regardless of how many ADCs are interleaved

Example (1)

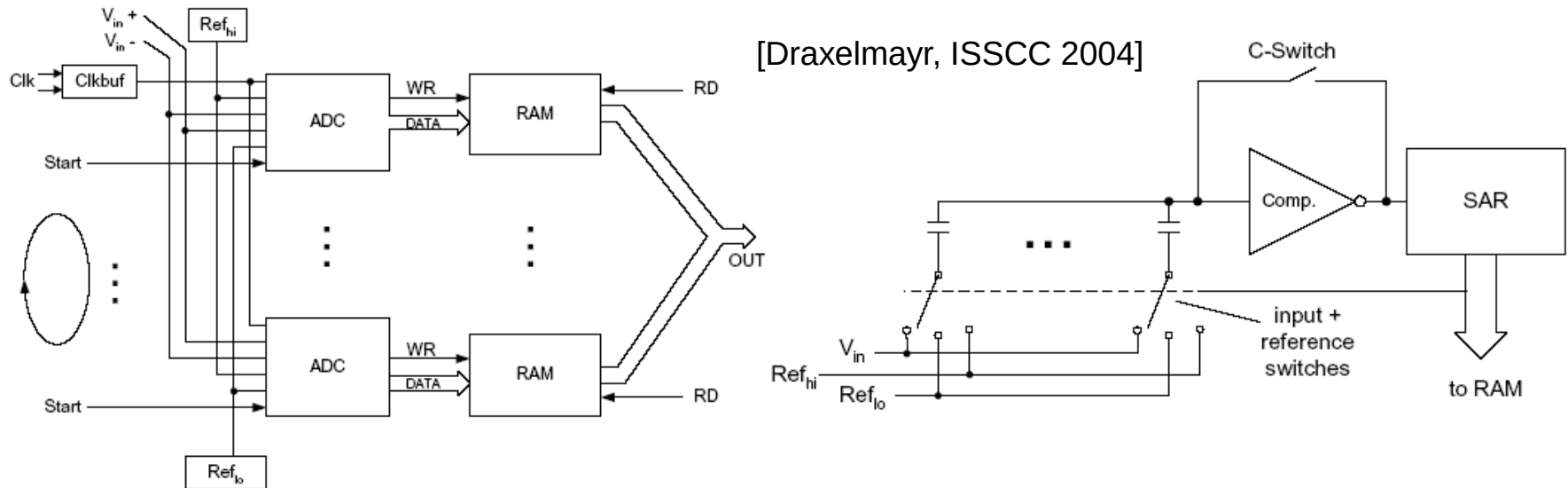


[Poulton, ISSCC 2003]



Sample Rate	20 GSa/s	
Resolution	8 bits	
INL Intrinsic With linearity correction	± 1.7 LSBs ± 0.4 LSBs	
DNL	± 0.3 LSBs	
Bandwidth	6.6 GHz	
Accuracy @ 500 MHz input @ 6 GHz input	6.5 effective bits 4.6 effective bits	
Jitter	0.7 ps rms	
Input Range	0.25 V _{pk} differential	
	Buffer Chip	ADC Chip
Input Capacitance	0.2 pF	4 pF
Power	1 W	9 W
Chip Size	1.2 x 2.6 mm	14 x 14 mm
Technology	40-GHz SiGe BiCMOS	0.18- μ m CMOS
Transistors	1000	50M
Package	438-ball BGA	

Example (2)

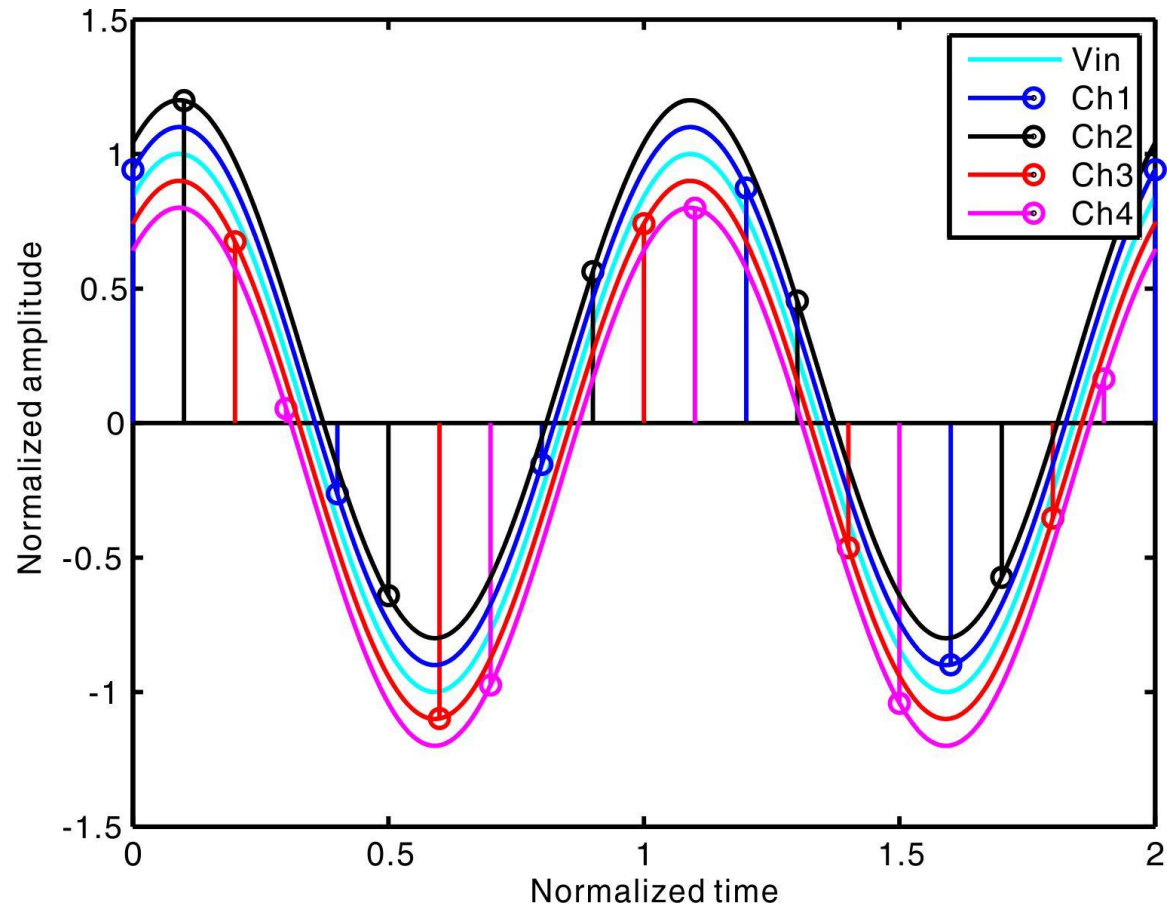


- Interleave "slow" SAR ADCs to get high throughput while maintaining low complexity and good power efficiency
- Eight 6-bit SAR ADCs, each running at 75 MS/s, overall throughput of 600MS/s, power=10mW in 90-nm CMOS technology

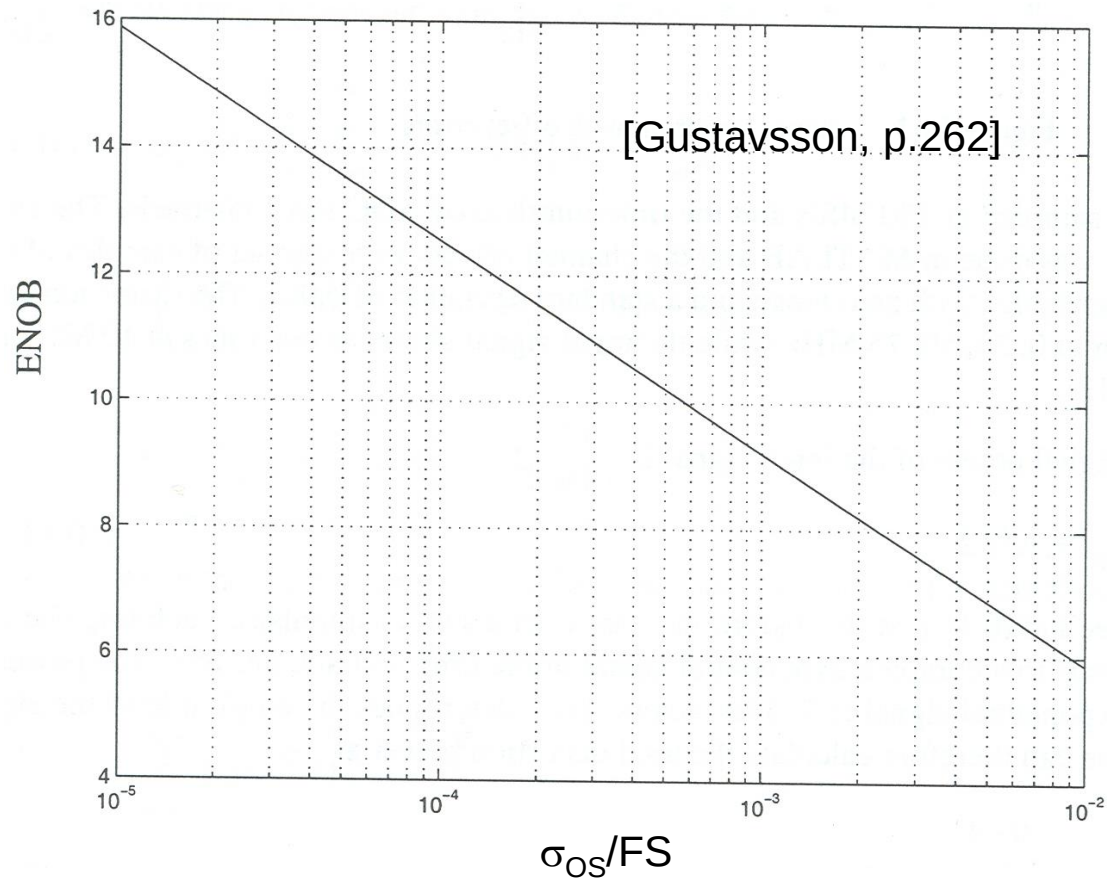
Issues with Time Interleaving

- Offset mismatch
 - Channels may have different offsets
 - Output will contain a periodic error sequence that manifest itself as spurs in the output spectrum
- Gain mismatch
 - Channels may also have different gains
 - Results in amplitude modulation
- Clock skew
 - Hard to guarantee precise phase relationship between individual clocks
 - Results in phase modulation (similar to aperture uncertainty)
- Solutions
 - "Careful design"
 - Analog or digital calibration
 - See e.g. [Jamal, JSSC 12/2002]

Offset Errors

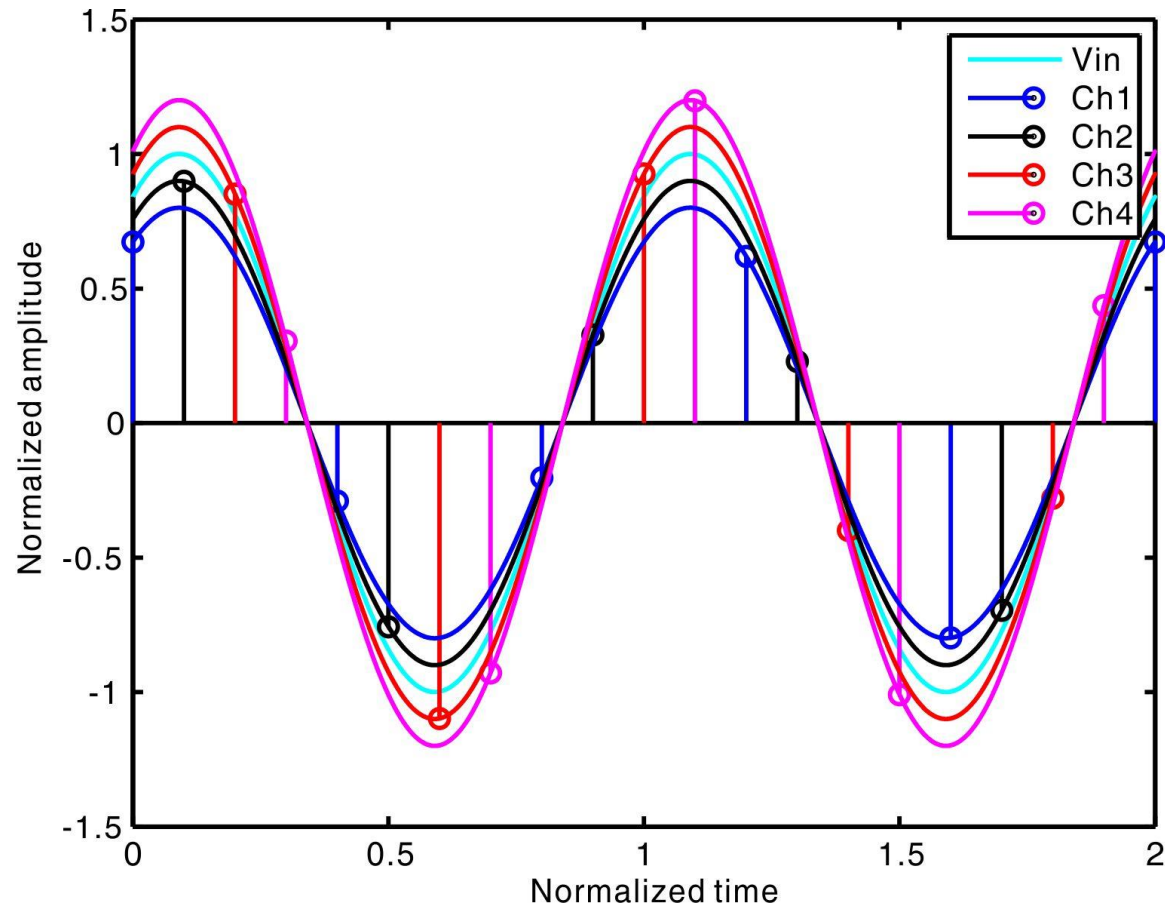


Impact of Offset Errors

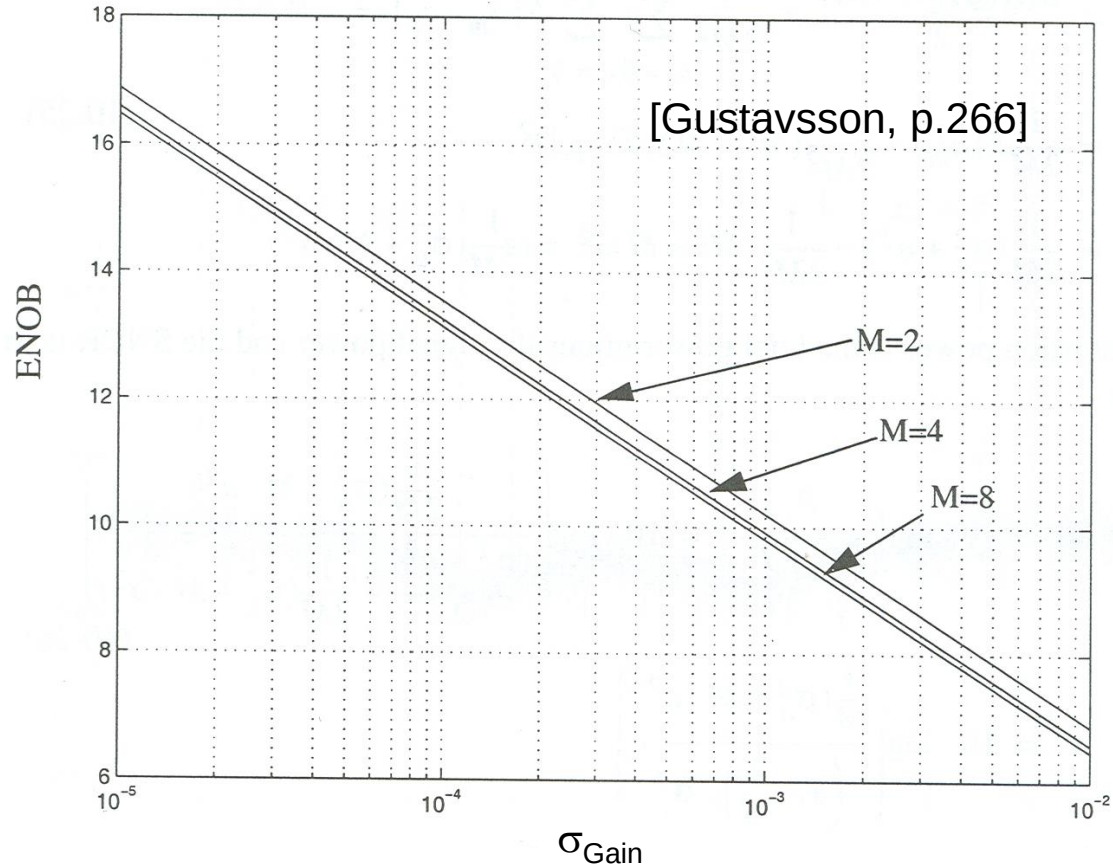


- E.g. $FS=1V$, $\sigma_{OS}=1mV \Rightarrow ENOB \sim 9\text{bits!}$

Gain Errors

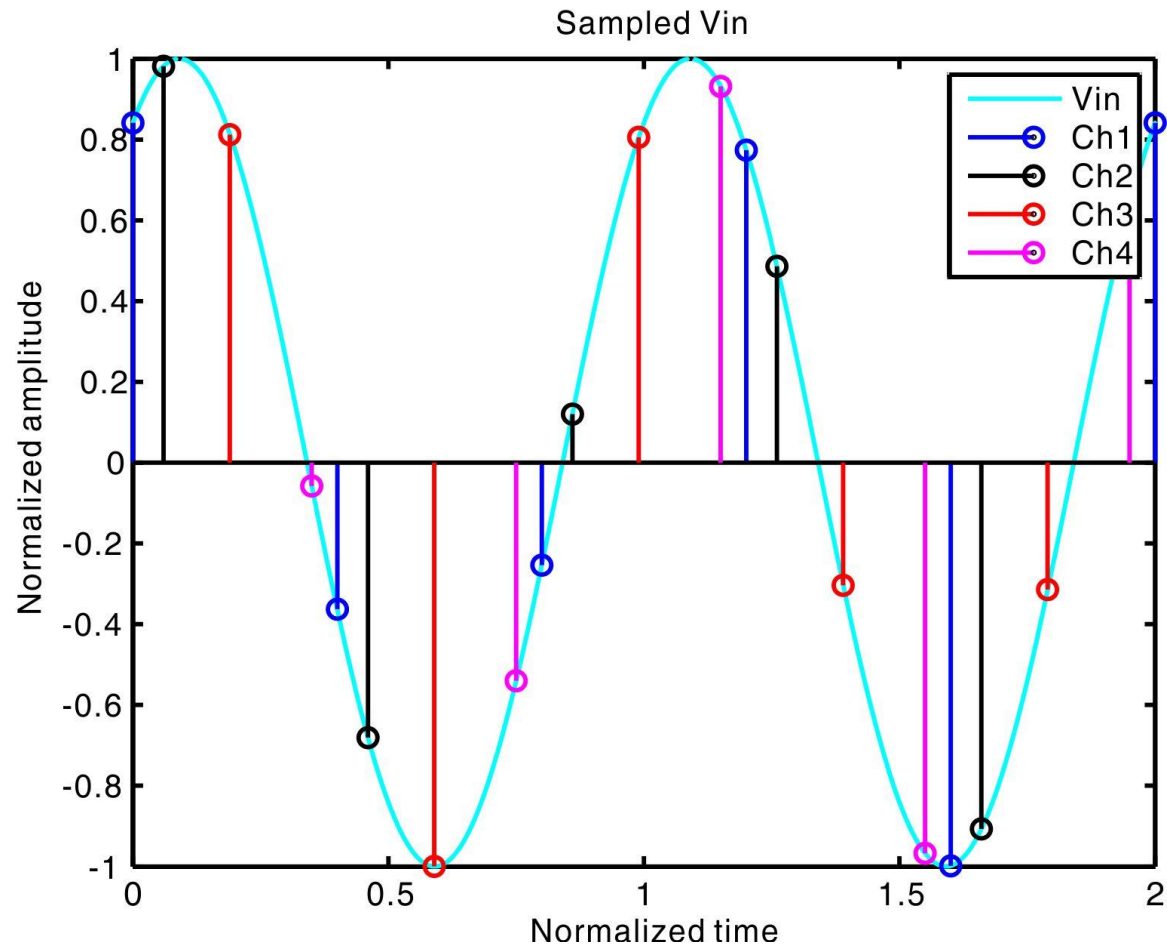


Impact of Gain Errors

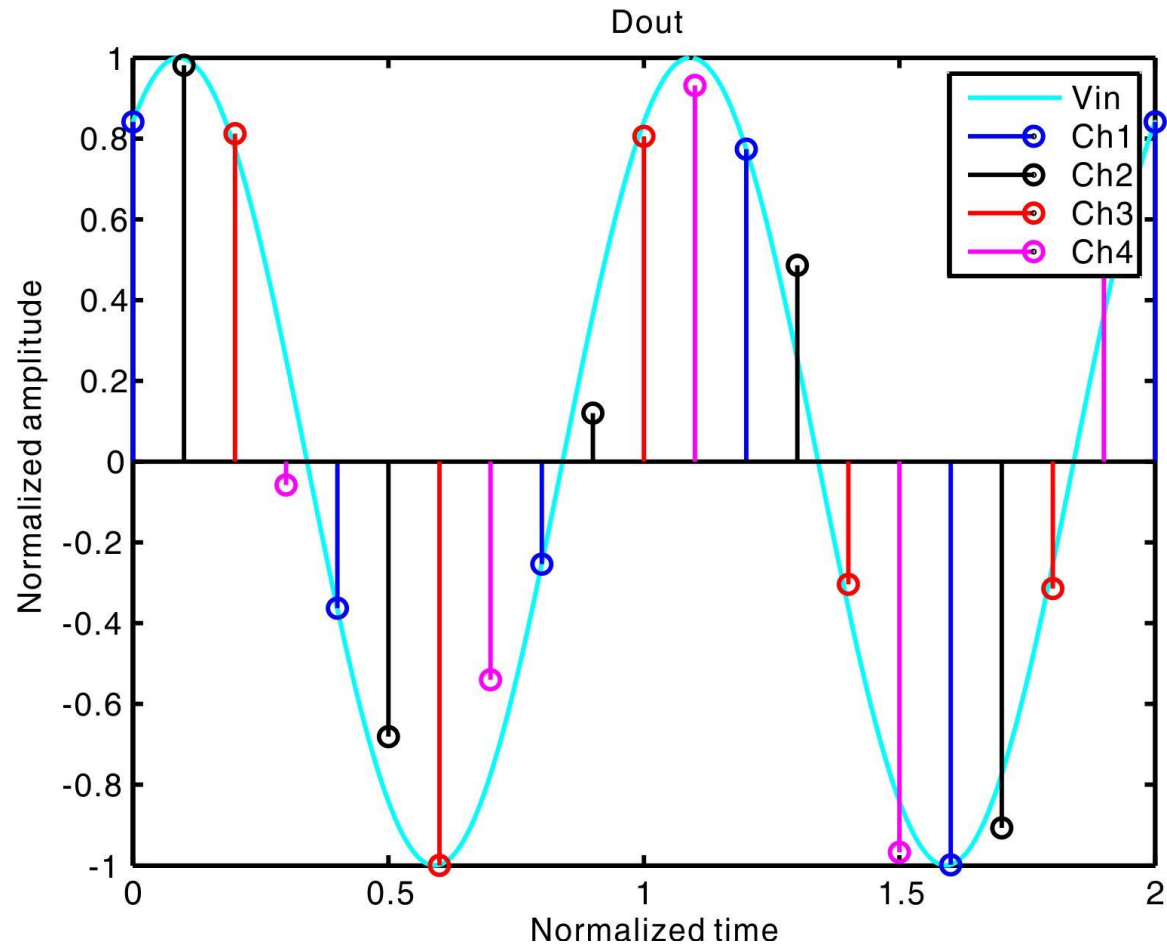


- E.g. $\sigma_{\text{Gain}} = 0.1\% \Rightarrow \text{ENOB} \sim 10 \text{ bits}$

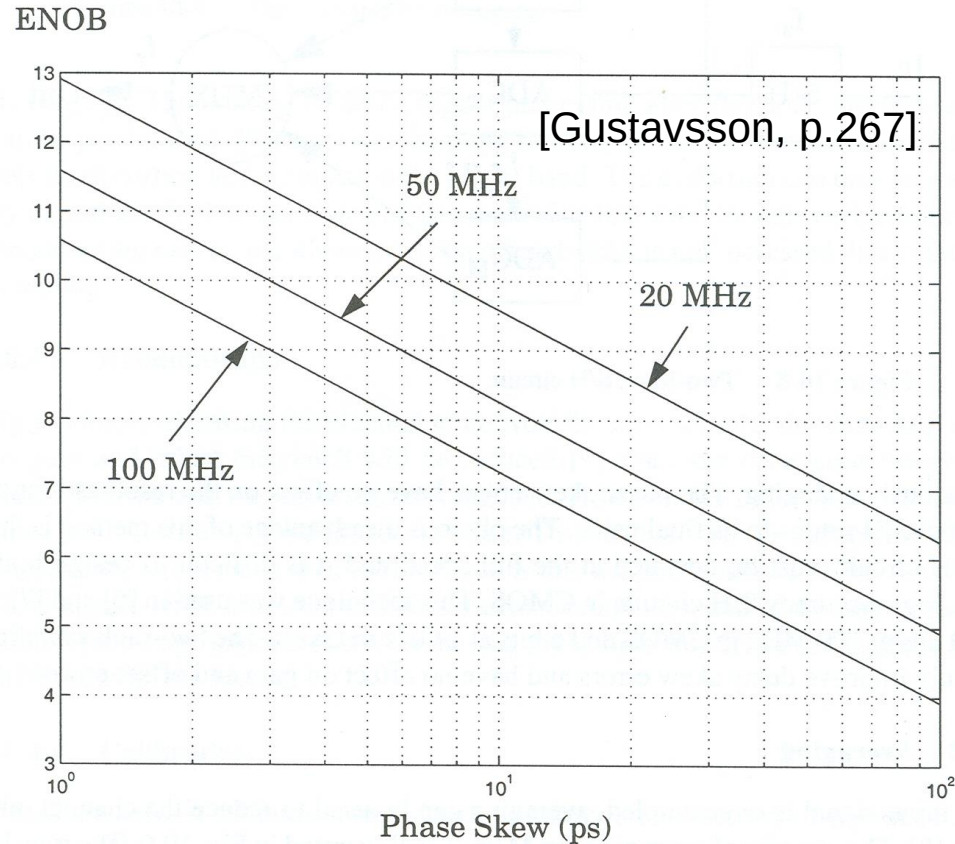
Clock Skew (1)



Clock Skew (2)



Impact of Clock Skew



- Above chart is for $M=4$ channels
- E.g. $f_{in}=100\text{MHz}$, phase skew=3ps $\Rightarrow$ ENOB~9bits!

Timing-Skew Estimation Techniques

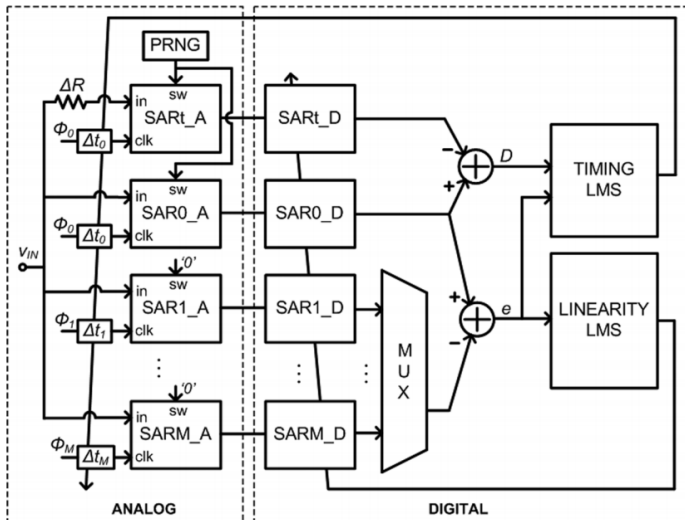
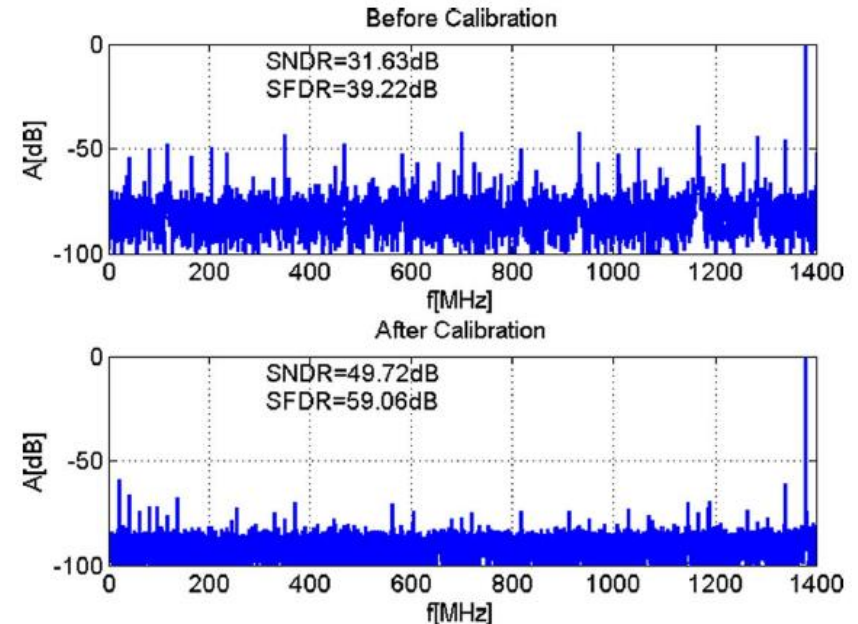
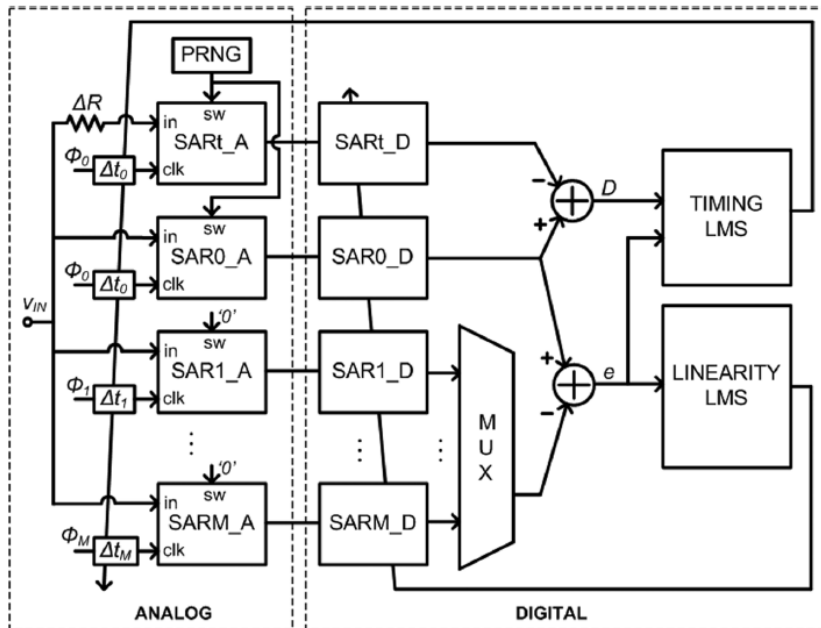


Image courtesy by
[Stepanovic, JSSC 2013]

- Timing mismatch estimation using two additional channels
 - Computes v_ϵ and $\frac{dV_{in}}{dt}$ to estimate Δt
 - Fast convergence speed
 - Spurs from input impedance variation
 - Large power and area overhead

Calibration Example

[Stepanovic and Nicolic, JSSC 2013]



- 24 8-bit SAR ADCs, aggregate throughput of 2.8GS/s, power=45mW in 65-nm CMOS technology

Prior Timing-Skew Calibration Works

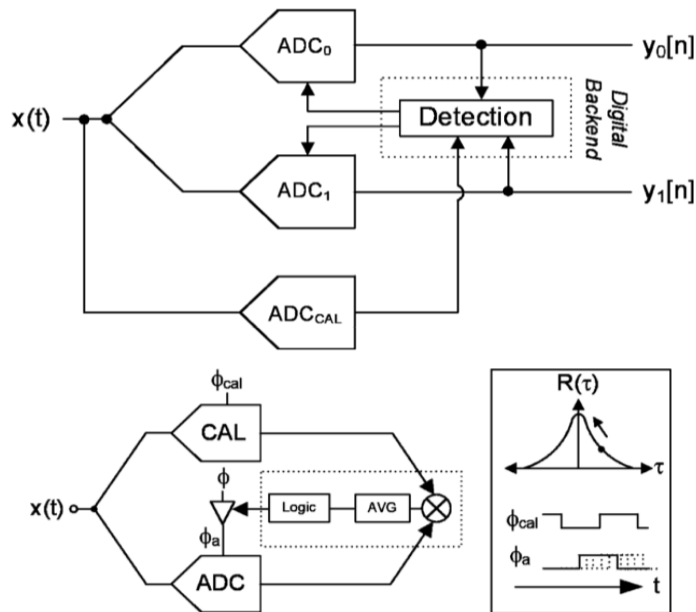
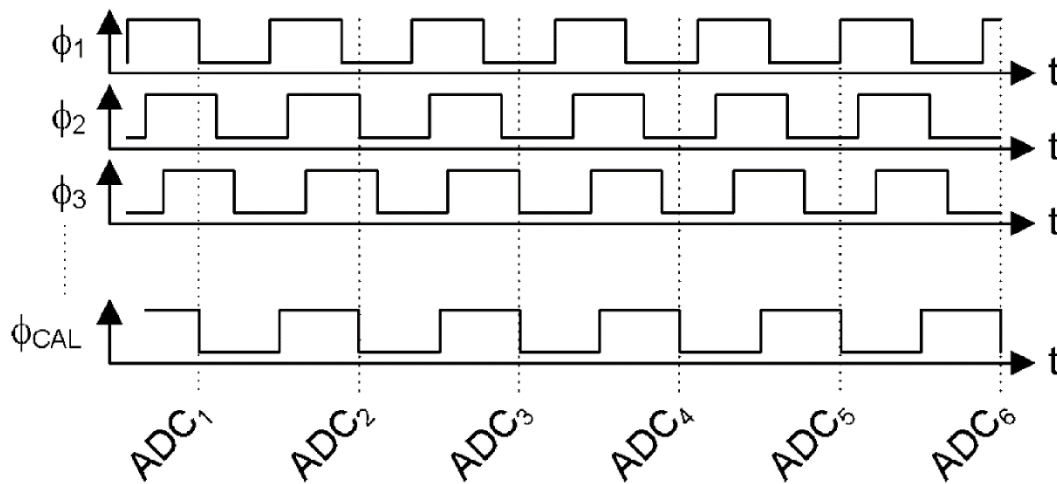
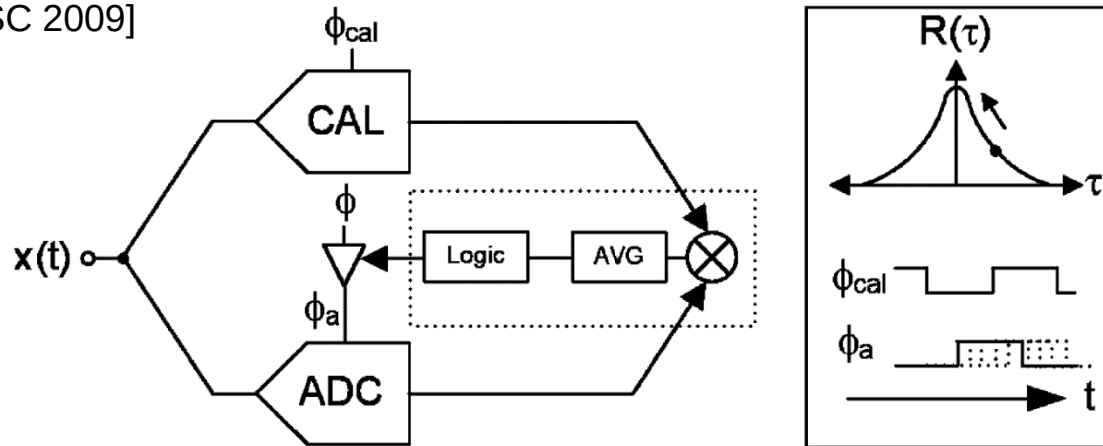


Image courtesy by
[El-Chammas, JSSC 2011]

- Timing mismatch estimation with a comparator-based reference
 - cross-correlation based estimation
 - Power-efficient reference
 - Slow convergence speed with random signals

Background Calibration of Clock Skew

[El-Chammas, JSSC 2009]



Timing-Skew Estimation Techniques

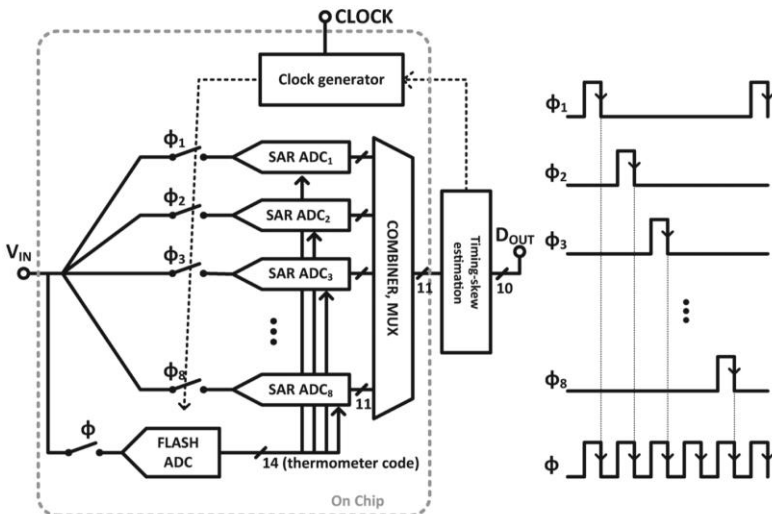


Image courtesy by
[Lee, JSSC 2014]

- Timing mismatch estimation with flash ADC as a reference
 - Variance-based estimation
 - No spurs from input impedance variation
 - Slow convergence speed with random signals
 - Power-hungry Flash ADC
 - Large variance computation power

Timing-Skew Estimation Techniques

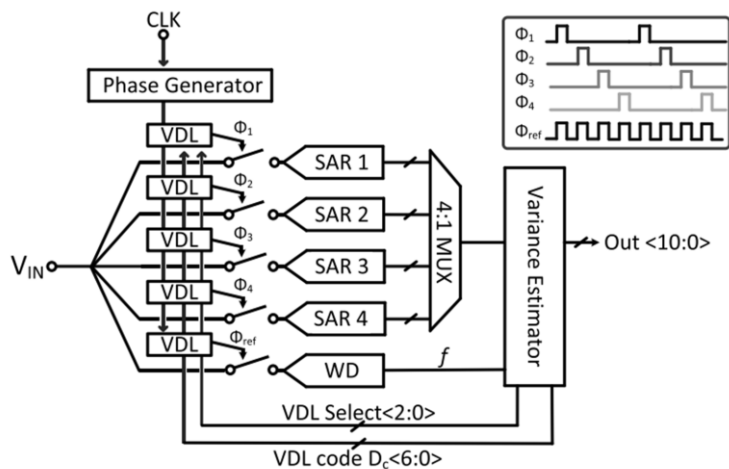


Image courtesy by
[Song, JSSC 2017]

- Timing mismatch estimation with a single comparator reference
 - Variance-based estimation
 - **No spurs** from input impedance variation
 - **Faster convergence** speed with random signal inputs
 - **Power-efficient** reference
 - **Reduced** variance computation power
 - **Still requires** digital multiplication

Timing-Skew Estimation Techniques

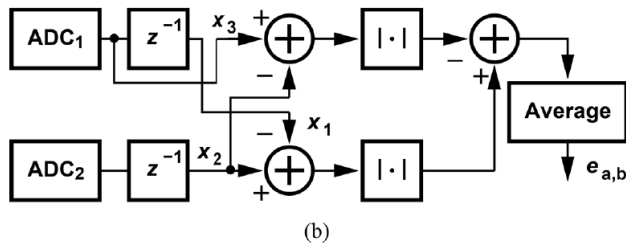
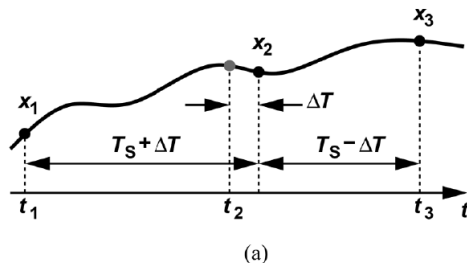


Image courtesy by
[Wei, JSSC 2014]

- Timing mismatch estimation without additional channel
 - Autocorrelation-based algorithm
 - Eliminated multiplication to reduce digital computation power
 - No extra hardware
 - Tight restrictions on the shape of input autocorrelation function

Block Diagram of the Proposed TI-SAR

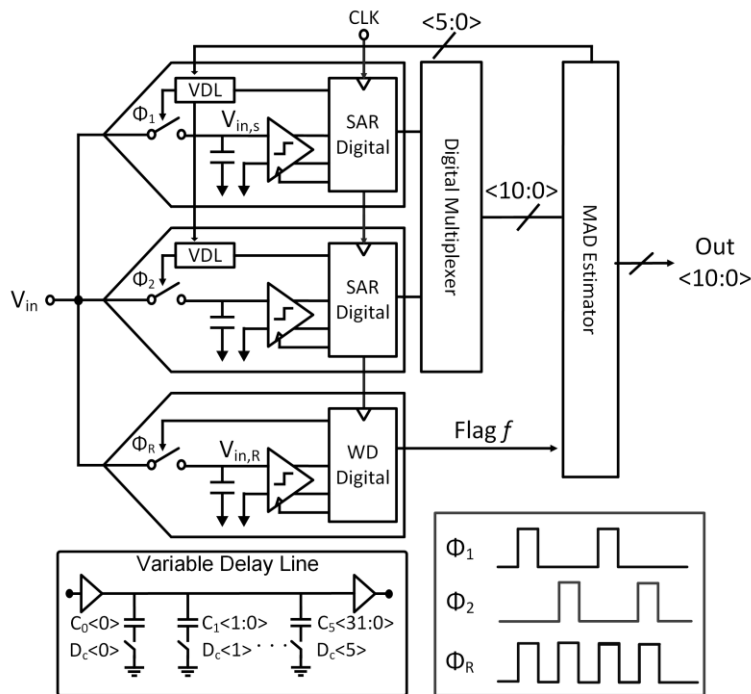


Image courtesy by
[Song, CICC, 2018]

- Two SAR ADCs
- Window detector reference
- Mean absolute deviation (MAD) estimator
 - $MAD = E[|D_{out}|]$
 - Uses MAD to estimate timing skew ($\Delta\tau$)
 - No multiplication required
 - Faster convergence speed even with random signals
 - no spurs from input impedance variation
 - Moderate input constraints

Fully digital background calibration

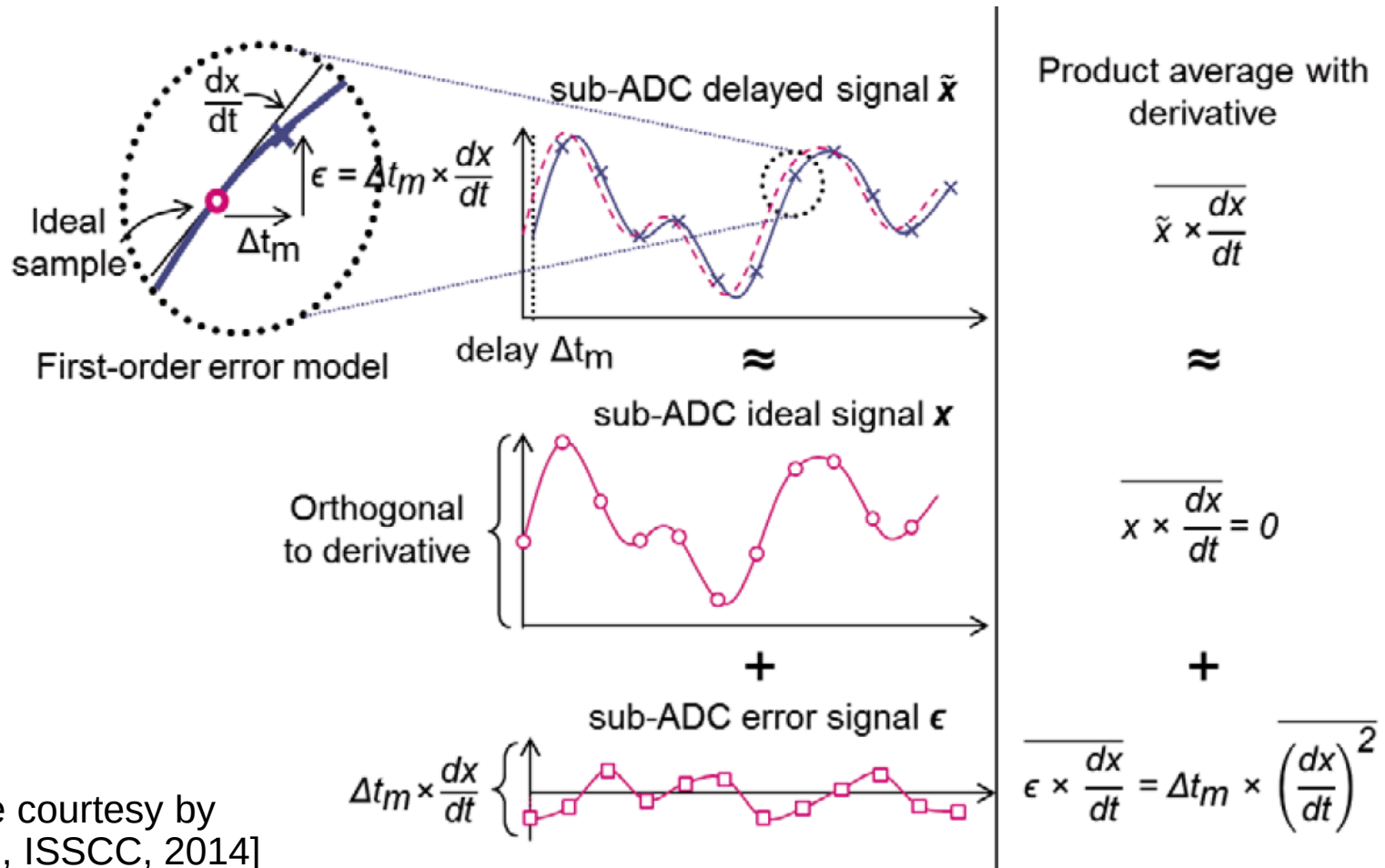
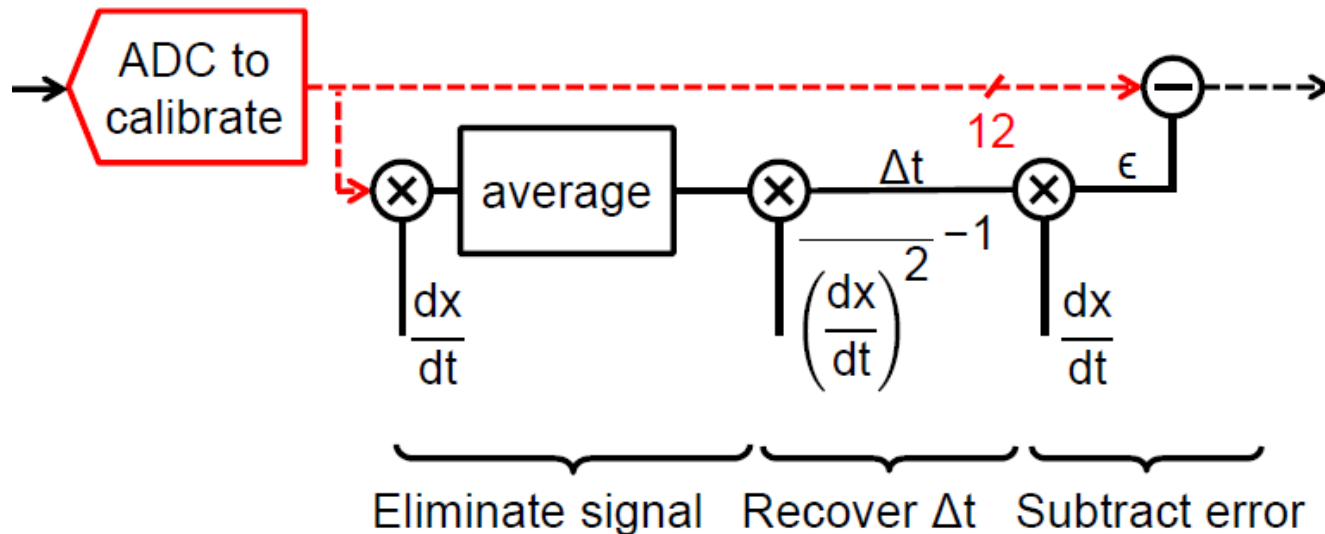


Image courtesy by
[Dortz, ISSCC, 2014]

Fully digital background calibration



No feedback loop in the calibration
→ No convergence issues

Image courtesy by
[Dortz, ISSCC, 2014]